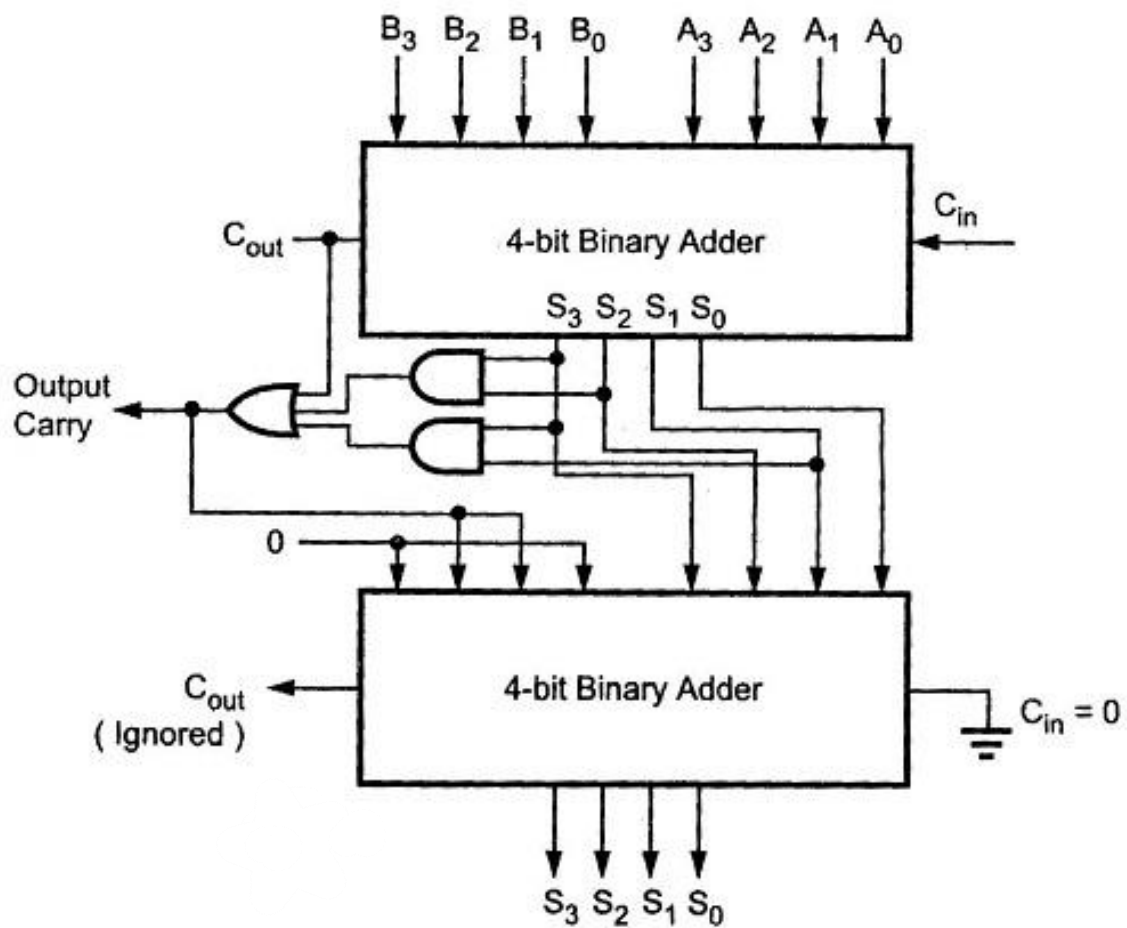


Day 22

4-bit BCD Adder in Verilog

Circuit Diagram :



Block diagram of BCD adder

Verilog Code

BCD Adder Verilog code

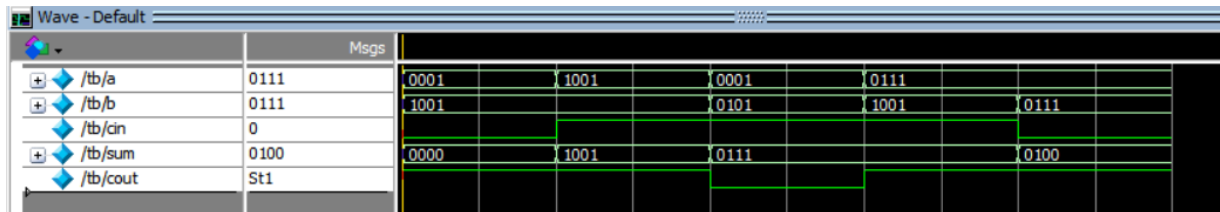
```
1 module bcd_adder(a,b,cin,sum,cout);
2   input [3:0]a,b;
3   input cin;
4   output reg [3:0]sum;
5   output reg cout;
6   reg [4:0]sum_temp;
7   always @(*) begin
8     sum_temp=a+b+cin;
9     if(sum_temp>9)
10    begin
11      sum_temp = sum_temp+6;
12      cout=1;
13    end
14  else begin
15    cout=0;
16  end
17  sum=sum_temp[3:0];
18  end
19 endmodule
```

Testbench Code

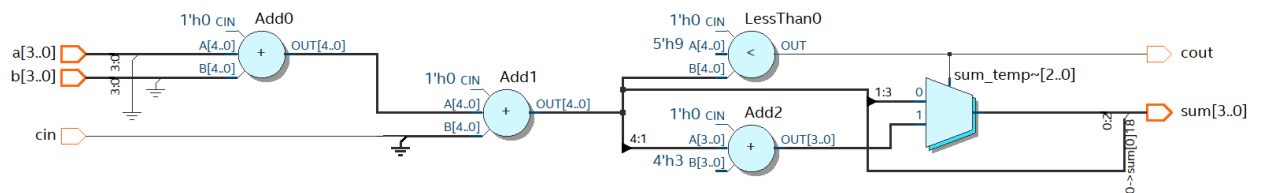
BCD adder Tb code

```
1 module tb;
2   reg [3:0]a,b;
3   reg cin;
4   wire [3:0]sum;
5   wire cout;
6   bcd_adder dut(a,b,cin,sum,cout);
7   initial
8   begin
9     a=4'b0001; b=4'b1001; cin=0; #10;
10    a=4'b1001; b=4'b1001; cin=1; #10;
11    a=4'b0001; b=4'b0101; cin=1; #10;
12    a=4'b0111; b=4'b1001; cin=1; #10;
13    a=4'b0111; b=4'b0111; cin=0; #10;
14    $finish;
15  end
16 endmodule
```

Waveform



Schematic



EDA Tools Used

- IntelQuartusPrime
- ModelSim